

|                  |    |                                                                                                                                                                               |
|------------------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dahua Camera     | 13 | Turn on, turn off, horizontal rotate, vertical rotate, video streaming, zoom in, zoom out, play voice, audio streaming, take photos, access the internet, update              |
| TP-Link DoorBell | 13 | Turn on, turn off, front camera, side camera, audio streaming, ring, infrared sensor, send a message, zoom in, zoom out, access the internet, informaton transmission, update |
| Eufy Doorbell    | 13 | Turn on, turn off, front camera, side camera, audio streaming, ring, infrared sensor, send a message, zoom in, zoom out, access the internet, informaton transmission, update |

**Table 4: Functionalities of Devices in Benchmark**

## B Baseline Prompts for Code Generator

### Naive Prompts

You are a code generator.  
Generate the code for {device\_brand} {device\_model}.

### Simple Prompts

You are a code generator.  
You can call Initial Searcher to retrieve relevant information about the device, which comes from Github Code, official API/SDK, official documentation/manuals, and developer blogs/websites.  
Generate the code for {device\_brand} {device\_model}.

### Improved Prompts

You are a code developer who can help me diversify the logic code of intelligent conversion into custom integration code for {device\_brand} {device\_model}.  
You can call Query the Catalog Tool with {device\_brand} {device\_model} to find the most relevant specification for it. If you have any misunderstanding, you can call Searcher Tool for futher retrieval.  
Once you believe that the code you provide is complete and executable, tell me that the final code has been implemented and show me the code file.  
The output Python code (custom integration code for {platform\_name}) allows me to use this code to control the {device\_brand} {device\_model} in {platform\_name}.  
Generate the code for {device\_brand} {device\_model}.

## C User Study: Participants' Demographic Characteristics and Subjective Feedback

Besides the last question in the self-designed questionnaire, we also conducted semi-structured interviews with participants to gather their thoughts and experiences in as much detail as possible. We summarize and analyze participants' subjective feedback as follows:

**Intuitiveness and User-Friendliness.** Participants felt that AutoBridge was simple to operate and highly integrated, effectively lowering the barrier to smart home development. *"It asks for very little input from me, and I don't need to understand any technical details—just follow its gentle prompts."* (N3) Most non-expert participants stated that AutoBridge is accessible. It enables smooth onboarding without requiring deep understanding, making it particularly beginner-friendly. *"No wasted time, no mental drain, super easy to pick up. If every smart-home app were straightforward like AutoBridge, I'd probably be a loyal customer."* (N2) Even for experienced developers, the simplicity of AutoBridge's interaction design was seen as a significant strength. *"It works just like the Cursor I use regularly – I only need to click 'yes/accept' or 'no/deny,' and the rest is taken care of automatically."* (E9) The high level of integration in AutoBridge minimizes the need for user intervention or supervision, allowing users to reallocate their time to other tasks.

**Developer Workload Mitigation.** The demand for automated code generation tools is not limited to novices; developers arguably need them even more. When writing integration code, developers often need to consult extensive documentation and strictly adhere to platform-specific coding standards, and AutoBridge effectively automates this effort. Expert-group participants felt that AutoBridge significantly reduces developer workload. *"The entire process of AutoBridge was remarkably convenient and efficient."* (E1) Given the wide range of branded and unbranded devices developers might encounter, managing integration across such diversity requires substantial time and expertise. AutoBridge's automation capabilities enable parallelism and greater productivity: *"If I were a hardware engineer, I'd probably face a wide*